Euro-Clear Ltd.
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MECHANIC GUIDE

INDUSTRIAL ACTIVATED CARBON ADSORBER EQUIPMENTS

WITH BLOCK HEAD CONTROL

BlueSoft 1054AT/67
BlueSoft 1354AT/67
BlueSoft 1465AT/67
BlueSoft 1665AT/67
BlueSoft 1865AT/67
BlueSoft 2162AT/75
BlueSoft 2472AT/75
BlueSoft 3072AT/75
BlueSoft 3672AT/111
BlueSoft 4272AT/111
BlueSoft 4872AT/111
BlueSoft 6386AT/112

Before using the equipment, read the guide carefully!

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1. DEFINITION AND FUNCTION

The equipment is a column filled with activated carbon filtrating material of excellent quality, which adsorbs carbon chloramines, chlorinated hydrocarbons and free active chlorine in treated water.

2. TECHNICAL PARAMETERS

Part number	Activated carbon liter	Control valve / Connection	Flow m3/h	Surface to filter m2	Backwash Flow	Weight kg	Size (mm) H x W x L
BlueSoft 1054AT/67	45	RX-67C-DTF 1″	0.5-1.0	0.05	1.2 m3/h	46	1600 x 300 x 270
BlueSoft 1354AT/67	72	RX-67C-DTF 1″	0.8-1.6	0.08	1.8 m3/h	62	1610 x 350 x 350
BlueSoft 1465AT/67	90	RX-67C-DTF 1″	1.0-2.0	0.1	2.3 m3/h	80	1890 x 370 x 370
BlueSoft 1665AT/67	112	RX-67C-DTF 1″	1.3-2.6	0.13	2.9 m3/h	96	1920 x 420 x 420
BlueSoft 1865AT/67	168	RX-67C-DTF 1″	1.8-3.6	0.18	4.0 m3/h	150	1940 x 500 x 500
BlueSoft 2162AT/67	224	RX-75A-DTF 2″	2.2-4.4	0.22	5.1 m3/h	190	1940 x 560 x 560
BlueSoft 2472AT/75	280	RX-75A-DTF 2″	2.8-5.6	0.28	6.5 m3/h	260	2170 x 620 x 620
BlueSoft 3072AT/75	392	RX-75A-DTF 2″	4.4-8.8	0.44	10.1 m3/h	430	2390 x 770 x 770
BlueSoft 3672AT/111	560	RX-111A-DTF 2″	6.4-12.8	0.64	14.7 m3/h	680	2500 x 940 x 940
BlueSoft 4272AT/111	784	RX-111A-DTF 2″	8.6-17.2	0.88	20.24 m3/h	920	2530 x 1210 x 1070
BlueSoft 4872AT/111	1008	RX-111A-DTF 2″	11.2-22.4	1.12	25.8 m3/h	1220	2510 x 1360 x 1220
BlueSoft 6386AT/112	1792	RX-112SM-DTF 2.5″	20-40	2.00	46.0 m3/h	2350	2510 x 1220 x 1360

Min. pressure during operation : 2,5 bar
Max. pressure during operation : 6 bar
Min. temperature during operation : 4 C°
Max. temperature during operation : 25 C°
Electrical connection : 230 V, 50 Hz

3. MAIN PARTS

The activated carbon filter mainly consists of the following parts:

3.1. Tank to filter

It is meant to store the catalytic charge. The columns are PE pressure tanks developed especially for treating water with polyethylene padding. Outside they have epoxy resin coat rolled by fibre.

Their features are that they last long, have little weight and resist to chemicals and corrosion.

3.2. Features of the charge of Activated carbon filter

Type	: Activated Carbon
Colour	: black
Density	: 0.44 kg/l
Grain size	: 0.6 – 2.3 mm
Iodin number:	: 950 or 1050 mg/g

The activated carbon's important functional properties:

Flow: 10-20 m/h
Rinsing volume flow rate: 22-28 m/h
Bed height: 900-1000 mm
Free surface: max. 50% of the bed height

3.3. Blocked valve to regenerate

It is meant to make sure that the mechanical operational processes of the equipment are completed in a programmed, regulated way.

Its parameters are given in the annex.

4. THE WAY IT WORKS

4.1. MECHANICAL OPERATION

During the equipment's operation the blocked valve's / control valve's mechanical operation ensures the automatic execution of the water production and the regeneration processes.

4.1.1. WATER PRODUCTION

During water production, the water gets into the column through the upper filter and it flows through the charge to remove carbon chloramines, chlorinated hydrocarbons and free active chlorine in treated water. The filtered water gets out of the equipment through the lower filter.

4.1.2. BACKWASH

During backwash the water gets into the column through the lower filter and it flows through the charge from down to up. Meanwhile, the filtrating charge gets stirred up. The water for backwash gets out into the channel through the drain.

4.1.3. DOWNFLOW WASHING

During the downflow washing the water gets into the upper filter into the column to store charge and it flows through the charge from up to down. It goes into the channel through the drain.During the downflow washing the recompression of the filtrating charge that has been stirred up is done.

4.2. DETAILED DESCRIPTION OF THE OPERATION CONTROL

The automatical blocked valve is solely time controlled during the operation and all the other working processes. There is an electronic clock for metering time. Apart from metering time, this clock gets the mechanisme of the blocked valve to work. The control unit has been programmed during production, the time for backwash is 2 o'clock at night. The backwash lasts for ~20 minutes. The frequency of backwash can be programmed between 1-99 days, depending on the system's needs.

5. INSTALLATION AND STARTUP

5.1. CONDITIONS OF INSTALLATION:

A room with flat, horizontal and hard flooring is needed for the installation of the equipment. The flooring and the direct surroundings of the equipment must resist to the corrosive effect of the brine. The equipment must be installed in a room the temperature of which is between +5°C to +40°C. The temperature of the raw water to be treated must not exceed +30 °C. The equipment must not be installed in a strongly damp or dusty room. It must be protected from frost, radiant heat and ultraviolet radiance.

Near the equipment, sewage connection and 230 V 50 Hz grounded, electric socket must be provided. In order to diminish risks, we advise to install the equipment into a room provided with floor drainage.

Below water pressure of 2,5 bars, proper regeneration is not guaranteed, so in this case we advise to build in a equipment to increase pressure.
In case the water pressure from the water system exceeds 6 bars, a equipment to decrease pressure has to be fixed in front of the equipment.

Fluctuation of pressure higher than ±0,5 bar is not allowed! Mechanical protective filter must be built in front of the equipment. It is important that the mechanical protective filter filtrates contamination bigger than 80 microns. The equipment does

not have any extra protection against water or electricity shortage. If needed, it must be provided while installing.

5.2. CONNECTING THE EQUIPMENT *(*it is the *customer's/owner's* job)

It is the customer's task to have the equipment connected to the water, sewage and electricity systems. The operator of the equipment and the specialist to complete the installation should both check if the equipment has been installed as described in the guide for use and handling and if the conditions to diminish risks of damage are given. The setting up of the equipment can be completed by the partner of the Euro-Clear Ltd's service that has a partnership contract. Setting up the equipment only means the setting up of the automatical control valve fulfilling the local conditions.

When setting up, the valid local regulations, general instructions and hygiene regulations must be followed and the technical parameters given above must be respected.

1. Pre-filtering equipment
2. By-pass branch
3. Equipment's Control valve
4. Drain
5. Electrical outlet
6. Gravitation drainage on the floor

The pre-filtering equipment No. 1 and the montageblock No. 2 are already fitted in when the customer gets them. The montageblock can be replaced by a by-pass branch built from 3 valves. When building in this unit into the pipe, make sure that you connect the raw water onto the filter „1" and the softened water that comes out has to be connected onto the montageblock „2".
The equipment and the montageblock can be connected by the flexible tube pair No. 3. When connecting, pay attention to the flow direction of the water which is marked by the arrows at the montageblock No 2 and the control valve No 4.

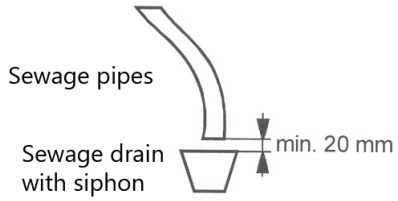
There is a hose outlet on the control valve No 4, which is the drain of the equipment. The water that comes out has to be led into the drain. This job can be completed by the plastic hose No 5. The hose must be pressure resistant as a simple garden hose breaks after a while and the narrow diameter can stop the completion of the regeneration. The sewage water comes out of the equipment under pressure, but it must be led by free outlet.

An electric plug underlaid of 230 V, 50 Hz must be built within a distance from the equipment that allows the completion of the connection of the prong plug number 6 without the electric cable's getting tight.

During backwash, water comes out from the pre-filtering equipment No 1. It is advised to connect the sewage connection snag into the drain. In this case the sewage water will come out under pressure, too.

The drain of the equipment has to be connected to the outlet points respecting the rules below.

- Respecting DIN 1988, the tube of the rinsing water and the overflow tube must be fixed at the sewage water connection point, at least at a distance of 20 mm compared to the highest sewage water level, so that the water can get out of the equipment smoothly.



5.3.SETTING UP OF THE EQUIPMENT

After the jobs mentioned in point 5.2 have been completed, you have to order the setting up of the machine from Euro-Clear Ltd at one of the contact details below:

Mailing address: 9071 Gönyű, Béke u. 2.
Mail address: contact@euro-clear.eu
Web address: www.euro-clear.eu

After the equipment has been set up, the copy of the warranty document filled in by the person having completed the setting up has to be sent to the address above in a verifiable way.

The fee of the setting up jobs gets calculated on the basis of the actual price list.

6. INSTRUCTION ABOUT HANDLING

The equipment is operated automatically, it does not need a permanent operator or inspection. Before or after a longer (for example 1 week long) outage, start the manual regeneration, cleaning process, and make sure that the equipment starts its operation under complete capacity.

1./ Check daily the pressure of the incoming water on the pressure gauge. In case the pressure decreases by more than 0,5 bars, clean the pre-filter.

2./ Clean the equipment with a dry cloth every two weeks.

3./The pre-filtering equipment must be rinsed back depending on the extent of the contamination but at least once a week. It must be done with the sewage water drain tap that you can find on the bottom of the equipment. Rinsing back must last for at least 15-20 seconds.

4./ Make sure that the equipment keeps getting 230 V, 50 Hz electricity all the time, for 24 hours and it is under system pressure continuously.

7. WARRANTY, GUARANTEE

In case of non-perfomance by the producer, the owner of the equipment can benefit from all warranty rights in 306-309. § in the Civil Code.

The owner of the equipment can claim for warranty and guarantee only by showing both the receipt that has been received when buying the equipment and that proves the payment of the complete price and the warranty document that has been filled in.

Warranty and guarantee do not cover faults that have been caused by the following:

- The product has not been used properly, the instructions of the way of handling, using, installing or maintaining etc. have not been respected
- The operation diary has not been kept
- the necessary corrective maintenance has not been completed, or has not been done by the designated professional servicing company,
- the product's nature has been transformed, changed
- the owner of the equipment has not completed their liability of reducing risks of damage
- Defects, damages and other problems caused by improper transportation and storage of the product.

Warranty and guarantee claims can be validated only in case the operator of the equipment sends to the producer both pages of the operation data sheet filled in and signed by the professional mechanic in charge of setting up of the equipment. It must be sent in a provable way.

Please send back to the address below both pages of the data sheet of setting up that have been filled in and signed:

Euro-Clear Ltd.
Mailing address: 9071 Gönyű, Béke u. 2.
E-mail: contact@euro-clear.eu

DATA SHEET FOR SETTING UP

Name of the specialist to complete setting up: …….

Contact details of the specialist completing the setting up:

- Mailing address: …….
- Telephone number: …….
- E-mail address: …….

Name of the company having sold the equipment: …….

Contact details of the company having sold the equipment

- Mailing address: …….
- Telephone number: …….
- E-mail address: …….

Name of the operator of the equipment: …….

Contact details of the operator of the equipment

- Mailing address: …….
- Telephone number: …….
- E-mail address: …….

Type of the equipment that has been set up: BlueSoft…….

Date of setting up: …….

.....
signature, stamp

Warranty and guarantee are only valid in case the setting up has been completed by Euro-Clear Ltd or its agent. You can order the setting up of the equipment at the contact details mentioned below.

Euro-Clear Ltd.
9071 Gönyű, Béke u. 2.
Tel: +3696/544-240
email: contact@euro-clear.eu

Data sheet for setting up

Yes

1. Check the mechanical and electricity connections as follows:
   - 1.1. Is a mechanical protection filter built in front of the equipment? ☐
   - 1.2. Is the pressure of the raw water convenient? (2,5 – 6 bars) ☐
   - 1.3. Are the directions of water flow convenient? (on the montage block, on the equipment) ☐
   - 1.4. Is the rinsing water outlet of the equipment connected into the channel? ☐
   - 1.5. Is the electricity input right? (230V, 50HZ) ☐

2. Program the control head of the filtrating equipment
   - 2.1. Have the exact date and time been set? ☐
   - 2.2. Setting the time for regeneration
   - 2.2 1. In case the equipment is time controlled, has the time between the two washings been set up? If yes, it is ……days
   - 2.3. Setting washing times (advanced settings) ☐
   - 2.3.1. Backwash (Backwash) Has the time been set up? ☐
   - 2.3.2. Rinse (Rinse) Has the time been set up? ☐

3. Starting manual wash, checking operation cycles:
   - 3.1. Backwash (water comes intensively into the channel).
     Is everything all right with the operation cycle? ☐
     After washing, is the water that comes out into the channel clean? ☐
   - 3.2. Post washing (bigger volume water into the channel) ☐
     After washing, is the water that comes out into the channel clean? ☐
   - 3.3. Repeat points 3.1. and 3.2. several times in a row as long as the outcoming water into the channel gets clear.
     Has the charge got clean? ☐
     How often did you have to repeat points 3.1. and 3.2. ? .....

4. Train the staff that handle the equipment. ☐

5. Filling in the warranty document ☐

WARRANTY DOCUMENT

In case the equipment is properly used, the producer undertakes a warranty of **12 months** starting from the setting up, but maximum **18 months** starting from the date of issuing the quality certificate.

DATE OF SETTING UP:

…….....

…….....
signature, stamp

The warranty and guarantee are only valid in case the setting up has been completed by Euro-Clear Ltd. or its agent. You can order the setting up of the equipment at the details mentioned below.

Please send us back the warranty document, setting up data sheet completely filled. In other case the warranty is not valid.

Please keep the warranty document, setting up data sheet and quality certificate for administration purposes in the future.

In case of a breakdown or fault, please inform us in written at the e-mail address contact@euro-clear.eu about the problem that has occured.

Euro-Clear Ltd.
Tel: +36 96 544-240 • E-mail: contact@euro-clear.eu • Web: www.euro-clear.eu

QUALITY CERTIFICATE

<table>
<tr><td colspan="2">1. Quality Certificate issued by:

Euro-Clear Ltd.</td><td colspan="2">2. Producer:

Euro-Clear Ltd.</td></tr>
<tr><td colspan="4">3. Punctual name of the product (its function):

Automatical activated carbon filter Type: BlueSoft ……</td></tr>
<tr><td>4. Quantity

1</td><td colspan="2">5. Weight and (or) size:</td><td>6. Date of production:

……</td></tr>
<tr><td colspan="2">7. Can be sold (used)</td><td colspan="2">8. Identifying product
a./ Control head number:
b./ ITJ-number: 36-10
c./ Part number:
d./ Other identifying details:</td></tr>
<tr><td colspan="2">9. Delivery and storage regulations:
Transportation and storage must be done in standing position. Store in a dry, cool place, away from water and precipitation. Do not expose to direct sunlight or UV radiation. Extremely frost-hazardous.</td><td colspan="2">10. Wrapping

Cardboard.</td></tr>
<tr><td colspan="4">11. Features of product (with punctual technical details, measurement results):

Volume flow: ……..……m^3/h

Volume of charge : ……..……litre

Quality and classifying: Convenient!</td></tr>
<tr><td colspan="4">12. Method of inspection for checking the quality of the product:

During production</td></tr>
<tr><td colspan="4">13. Regulations for use and handling:

As mentioned in the guide for use and handling</td></tr>
<tr><td colspan="2">14. Other details:</td><td colspan="2">15. Signature of the person to issue the quality certificate:

Gönyű, 20……

……
signature, stamp</td></tr>
</table>